# EXPERIMENT 2

**EXPERIMENT 3**

**Aim:** To Perform Various GIT Operations on Local and Remote Repositories using GIT Cheat-Sheet

**Theory:**

Git provides a wide range of operations to manage both local and remote repositories efficiently. A GIT Cheat-Sheet is a quick reference guide that lists all important Git commands and their usage. These operations include initializing a repository, staging and committing files, viewing logs, pushing and pulling from remote repositories.

**Important GIT Commands from Cheat-Sheet:**

git config — Configure user details git init — Initialize a new local repository git add — Stage files for commit git commit — Save staged changes to local repository git log — View commit history git status — Check current state of repository git push — Upload local commits to remote repository git pull — Download changes from remote to local git remote — Manage remote repository connections git branch — List or create branches git merge — Merge branches together

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**Practical Steps:**

**Operation 1: Configure Git (git config)**

Before performing any Git operation, user details must be configured. This is done using the git config command.

git config --global user.name "kunalmotwani223" git config --global user.email "kunalmotwani223@gmail.com" git config --global --list

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL
$ git config --global --list
user.name=kunalmotwani223
user.email=kunalmotwani223@gmail.com
```

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**Operation 2: Initialize Local Repository (git init)**

A new local repository is created using the git init command. This creates a hidden .git folder inside the project directory which tracks all changes.

mkdir git-demo-project cd git-demo-project git init

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-demo-project
$ git init
Initialized empty Git repository in C:/Users/Lenovo/Desktop/KUNAL/git-dvcs/git-d
emo-project/.git/
```

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**Operation 3: Stage and Commit Files (git add, git commit)**

Files are added to the staging area using git add and then permanently saved to the local repository using git commit.

git add . git commit -m "First Commit"

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/
mo-project (master)
$ git commit -m "First Commit"
[master (root-commit) 34ec911] First Commit
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 index.html
 create mode 100644 teststatus
```

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**Operation 4: View Commit History (git log)**

The git log command displays the history of all commits made in the repository. The --oneline option shows each commit in a single line for quick reference.

git log --oneline

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-de
mo-project (master)
$ git log --oneline
3b6afd6 (HEAD -> master) Express Commit
34ec911 First Commit
```

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**Operation 5: Connect and Push to Remote Repository (git remote, git push)**

To upload the local repository to GitHub, first a remote connection is established using git remote add, then changes are pushed using git push.

git remote add origin https://github.com/kunalmotwani223/SEPM_72.git git push -u origin master

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-demo-project (master)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 28 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 503 bytes | 503.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kunalmotwani223/SEPM_72.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

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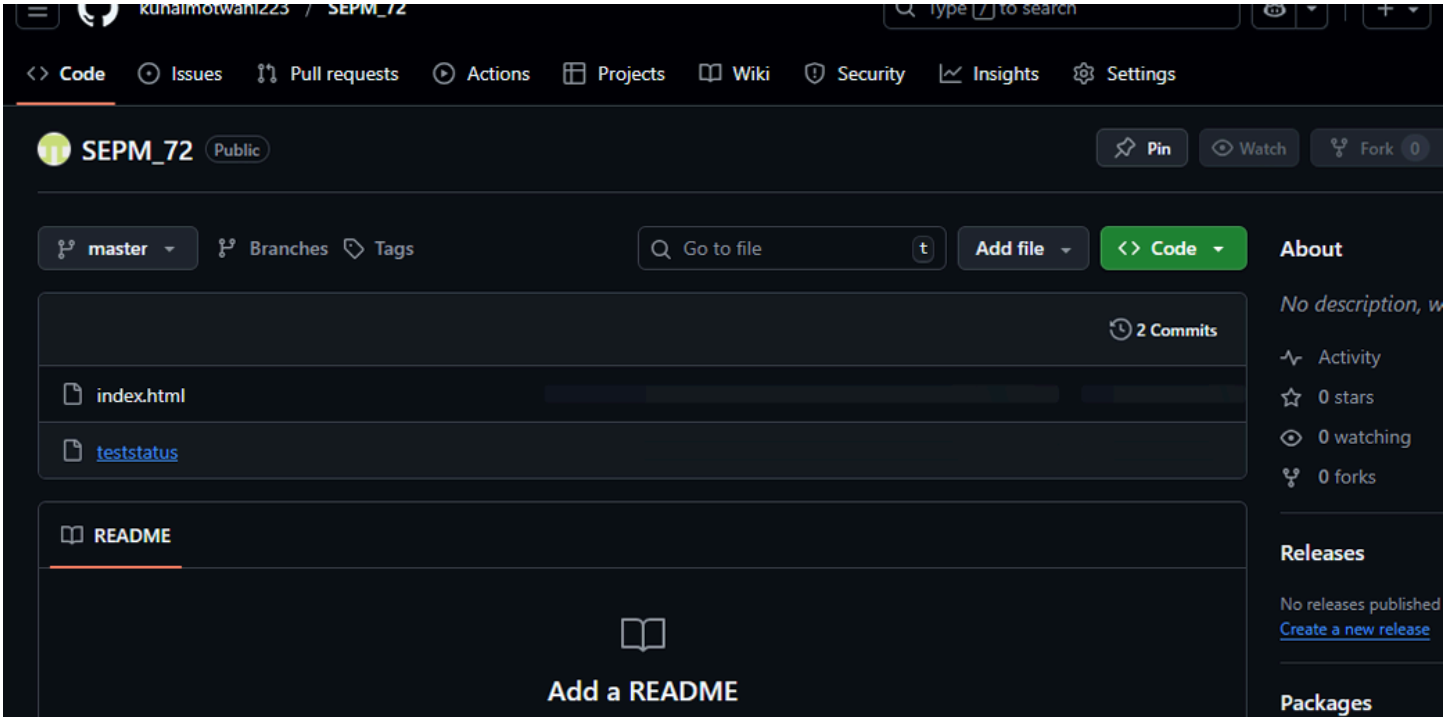
**Operation 6: Verify Remote Repository**

After pushing, all committed files are visible on the GitHub remote repository page. This confirms successful synchronization between local and remote repositories.

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## Operation 7: Push Updated Changes to Remote (git push)

Whenever new changes are committed locally, they can be pushed to the remote repository again using git push.

git push

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-demo-project (master)
$ git push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 28 threads
Compressing objects: 100% (1/1), done.
Writing objects: 100% (2/2), 249 bytes | 249.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kunalmotwani223/SEPM_72.git
   3b6afd6..78a3190  master -> master
```

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## Conclusion:

Various GIT operations on local and remote repositories were successfully performed using the GIT Cheat-Sheet as reference. Operations including git config, git init, git add, git commit, git log, git remote, and git push were executed and

verified. The GIT Cheat-Sheet proved to be a useful quick reference for performing these operations efficiently.